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Lab 3 – Written Justification

Problem Statement #29: Municipal Infrastructure Damage Reporting App

Architecture Selection: “We chose Microservices Architecture for the Municipal Infrastructure Damage Reporting App.”

Architectural Choice

We went with Microservices Architecture. Citizen Reporting App, Report Intake & Photo Processing Service, GIS Location Service, Ticket Dispatch Service, Ticket Data Store, and Dept. Notification & Engineer Portal are each their own independently deployable service, talking to each other only through the interfaces shown in the diagram below. We looked at a Layered approach too, but it didn't fit as well once we noticed how differently each responsibility behaves under load — photo processing spikes, dispatch is constant, and the engineer portal is fairly steady.

Reason 1 – Department Separation

Road Infrastructure, Water, and Electrical departments are separate downstream consumers of a ticket. Keeping ticket dispatch as its own service means the routing rules for one department can be changed without tightly coupling the other departments' workflows.

Reason 2 – Bursty Workload

Photo upload and EXIF/GPS extraction can see sudden spikes right after events like heavy rain, while the Engineer Portal usually sees fairly steady traffic. Separating these responsibilities lets the Report Intake & Photo Processing Service scale independently instead of the whole system having to scale together.

Security Advantage

The Citizen Reporting App never talks to the Ticket Data Store directly — it only goes through the Report Intake service's public API. That reduces direct exposure of ticket and department data and keeps dispatch/data access behind service boundaries instead of reachable straight from the client app.

Performance Benefit

The Report Intake & Photo Processing Service can be scaled horizontally during peak reporting periods (like right after a storm) without scaling every other service. This avoids over-provisioning services like the Engineer Portal that don't need it, while keeping photo and GPS processing responsive when reports come in fast.

Component Diagram

6 components inside the system boundary (Citizen Reporting App, Report Intake & Photo Processing Service, GIS Location Service, Ticket Dispatch Service, Ticket Data Store, Dept. Notification & Engineer Portal). 6 interfaces shown with ball (provided) / socket (required) notation, each labelled with its protocol and payload.

